

Activity Wheel!

Feature Overview

A gamified activity suggestion app for young children

Built with React + Vite | Deployed on GitHub Pages

AI-powered by Claude via Cloudflare Workers

`mattcinnamon/activity-wheel`

<https://mattcinnamon.github.io/activity-wheel/>

1. Overview

Activity Wheel! is a single-page web app designed to help parents and carers overcome decision fatigue when planning activities with young children (3+ years). Children spin an apple-shaped wheel to land on one of six themed categories, then receive a fun, age-appropriate activity to do with their grown-up.

The app features **66 hand-crafted default activities** across six categories, with the ability to generate additional personalised activities using AI. Progress is tracked per category, and a star reward system provides extra motivation.

2. Core Gameplay Loop

The main user flow follows these steps:

- **Spin the Wheel** - Tap the apple-shaped wheel on the homepage to spin it.
- **Category Selection** - The wheel lands on one of 6 activity categories (or a bonus star).
- **Envelope Reveal** - A 3D envelope flips open to tease the activity.
- **Mission Card** - The full activity title and description are shown with action buttons.
- **Complete or Shuffle** - Mark the activity as done, shuffle for a different one, or spin again.
- **Progress Tracking** - Completed activities are logged with timestamps and shown on the homepage.

If a star is landed on, a celebration animation plays with confetti, and the star count increments. Reaching 10 stars triggers a special reward message.

3. Activity Categories

The app includes six themed categories, each with a unique emoji, colour, and set of activities:

Explore (Magnifying Glass) - 12 activities

Colour: #4CAF50

Outdoor observation and nature discovery: leaf hunts, harbour adventures, bug safaris, cloud spotting, colour walks, puddle patrol, texture exploring, sound detecting, shadow catching, rock collecting, flower spotting, bridge counting.

Experiment (Test Tube) - 12 activities

Colour: #2196F3

STEM and science activities: volcano fizz, magic milk, sink or float, rainbow walks, bread science, ice rescue, raisin dancers, magnet exploring, colour mixing, seed growing, sound shakers, static hair.

Dance (Dancer) - 10 activities

Colour: #E91E63

Movement and music activities: animal dance party, freeze dance, scarf dancing, copy cat moves, bubble dance, slow-mo disco, musical feelings, dance stories, shadow dance, rhythm maker.

Puzzle (Jigsaw) - 10 activities

Colour: #FF9800

Logic and pattern activities: shape hunting, matching pairs, pattern building, sorting challenges, treasure maps, what's missing, counting quests, story sequencing, mirror puzzles, odd one out.

Build (Brick) - 10 activities

Colour: #795548

Construction and making activities: tallest tower, animal house, bridge builder, castle creator, marble run, robot friend, train track town, blanket fort, boat builder, magnetic shapes.

Paint (Palette) - 12 activities

Colour: #9C27B0

Art and creativity activities: handprint animals, blow painting, nature art, bubble wrap stamping, colour mixing magic, self portrait, sponge painting, rainy day art, playdough creations, collage creation, pasta jewellery, finger painting feelings.

Bonus: Star Segment

The wheel includes a star segment (half the angular size of other segments, lower probability). Landing on it triggers a celebration with confetti animation and increments the star counter. Collecting 10 stars displays a special reward message.

4. Animations & Visual Effects

Apple Wheel

- Apple-shaped SVG wheel with clip path, stem, and leaf
- 4-second spin with cubic-bezier easing for natural deceleration
- Weighted random selection: star has half the probability of other categories
- Text labels radiate outward from centre; category emoji at outer edge

Caterpillar Eating Animation

- After the wheel stops, a caterpillar drops in with bounce animation
- Three progressive bite stages (small, medium, large) with growing bite marks
- Bite appears at the selected segment's midpoint via SVG mask
- Wheat-coloured circle shows apple 'flesh' beneath the bite
- 2-second delay before transitioning to the envelope

Envelope 3D Flip

- Large category emoji displayed on the envelope front face
- On tap: envelope flips 180 degrees via CSS 3D transform (0.6s)
- Envelope flap opens upward after 600ms
- Mission card slides out of envelope after 1200ms
- Transitions to full mission card view after 1800ms

Celebration (Star)

- Star emoji spins continuously with scale pulsing
- 60 confetti particles fall with random colours, sizes, and rotation
- Auto-closes after 5 seconds (tap to dismiss early)
- Reward bounce animation when 10 stars reached

5. Screens & Navigation

The app uses a phase-based state machine (no router) with seven phases:

- **Homepage** - Main landing with title, live mission card, tiles, and progress grid
- **Wheel** - Apple-shaped spinning wheel with star counter
- **Envelope** - 3D flip reveal animation for selected category
- **Mission Card** - Activity details with Done, Shuffle, and Spin Again buttons
- **Celebration** - Star reward animation with confetti
- **Completed** - All finished activities grouped by category with dates
- **Menu** - Settings with sub-pages for Names, Activities, and Clear History

6. Live Mission Persistence

When a mission is selected, it becomes the 'active mission' and persists across page reloads via `localStorage`. The homepage displays a prominent Live Mission card showing:

- Category emoji and colour
- Activity title (with 'Custom' badge if AI-generated)
- Time elapsed since started (e.g., '2h ago', 'Just now')
- Activity description (personalised with saved names)
- 'Mark Complete' and 'View' action buttons

7. Settings Menu

Accessible via the burger menu emoji fixed in the top-right corner of the homepage.

Names

- **Little Person** - Child's name for personalisation
- **Big Person 1** - Primary parent/carer name
- **Big Person 2** - Optional second parent/carer name
- Save button with visual confirmation feedback

Edit Activities (AI Generation)

- Criteria form: age, likes & interests, dislikes/sensitivities, location, nearby places
- 'Generate Custom Activities' button sends criteria to Cloudflare Worker
- AI generates 3 tailored activities per category (18 total)
- Generated activities displayed below, grouped by category
- Each activity has Edit and Delete buttons for manual adjustment
- Custom activities are merged into the main wheel pool alongside defaults
- Inline editing with editable title and description fields

Clear History

- Warning screen listing all data that will be reset
- Clears: completed missions, star count, active mission, names, custom activities
- Requires explicit confirmation via danger button

8. Personalisation

Name Substitution

Activity descriptions containing 'Big Person' are dynamically personalised with the saved names from Settings. The replacement is applied everywhere descriptions are shown: mission cards, homepage live mission, and completed activities.

- **Both names set:** '...your Big Person, Matt or Jess, ...'
- **One name set:** '...your Big Person, Matt, ...'
- **No names:** Text remains unchanged

Custom Activity Badge

AI-generated activities display a purple 'CUSTOM' badge next to the title, appearing on the mission card and homepage live mission. This helps the big person quickly identify which activities are custom vs. from the original set.

9. AI-Powered Activity Generation

Architecture

The app uses a Cloudflare Worker as a secure proxy to the Claude API (Anthropic). This keeps the API key server-side and adds rate limiting and origin checking.

- **Model:** Claude Sonnet (claude-sonnet-4-20250514)
- **Worker URL:** <https://activity-wheel-api.mcbcinnamon.workers.dev>
- **Rate Limit:** 10 requests per hour per IP address
- **Origin Check:** Only accepts requests from mattcinnamon.github.io
- **Storage:** Cloudflare KV for rate limit tracking

Security

- API key stored as Cloudflare Worker secret (never exposed to frontend)
- CORS restricted to the GitHub Pages origin
- IP-based rate limiting prevents API credit abuse (10 req/hr)
- Request validation: POST-only, JSON body required, criteria field required

How It Works

1. User fills in criteria (age, interests, location, etc.) in the Edit Activities page.
2. Tapping 'Generate' sends criteria + saved names to the Worker.
3. Worker builds a prompt and calls the Claude API.
4. Claude returns 3 activities per category (18 total) as JSON.
5. Activities are saved to localStorage and appear in the custom activities list.
6. Custom activities are merged into the wheel's mission pool for future spins.

10. Data Storage

All data is stored in the browser's localStorage under a single key ('activity-wheel-state'). No user accounts or server-side database required. Data persists across sessions on the same device.

- **Completed Missions:** Array of objects with category key and ISO timestamp
- **Star Count:** Integer counter for bonus stars collected
- **Active Mission:** Current in-progress mission with category, title, description, timestamps
- **Names:** Little person + up to 2 big people names
- **Custom Activities:** AI-generated activities keyed by category
- **User Criteria:** Saved form inputs for AI generation

The app includes automatic data migration: older localStorage formats (string arrays) are transparently upgraded to the current object format with timestamps.

11. Technology Stack

- **Frontend:** React 19 (hooks only, no Redux or Router)
- **Build Tool:** Vite 8
- **Hosting:** GitHub Pages (via GitHub Actions CI/CD)
- **AI Backend:** Cloudflare Workers (free tier)
- **AI Model:** Claude Sonnet (Anthropic API)
- **Rate Limit Store:** Cloudflare KV
- **Styling:** Vanilla CSS with 3D transforms and animations
- **Font:** Nunito (Google Fonts)
- **Repository:** github.com/mattcinnamon/activity-wheel

12. Deployment

Frontend (GitHub Pages)

- Automatic deployment on push to main branch via GitHub Actions
- Node 20, npm ci, vite build, upload to GitHub Pages
- Base path configured as /activity-wheel/

Backend (Cloudflare Worker)

- Deployed separately via Wrangler CLI from the /worker directory
- API key stored as encrypted Cloudflare secret
- KV namespace for rate limiting state
- No ongoing maintenance required - runs on Cloudflare free tier